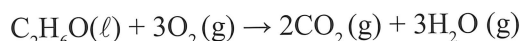


Set 29: Stoichiometry

Notes

A driver has been stopped by the police as part of a random breath test. There are a number of analytic techniques that can be used to determine the amount of alcohol in the driver's bloodstream. Many of these techniques rely on reacting the alcohol with other chemicals and measuring the amount of products formed. In order to determine the amount of alcohol an understanding of the relationship between the reactants (including alcohol) and the products in the reaction is required. This relationship, expressed with the aid of a chemical equation, is called stoichiometry.

For example the equation representing the burning of ethanol is:



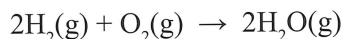
This indicates that one mole of $\text{C}_2\text{H}_6\text{O}$ reacts with three moles of O_2 to form two moles of CO_2 and three moles of H_2O .

Using reaction stoichiometry, the relationship between the number of moles, masses and gaseous volumes of reactants and products can be calculated. We will start with the relationship between masses and moles.

Examples

1. When a mixture of hydrogen and oxygen is sparked an explosive reaction occurs in which water is formed. If 0.480 moles of hydrogen gas is mixed with excess oxygen and sparked, calculate:
 - (a) the number of moles of water formed
 - (b) the mass of water formed
 - (c) the number of moles of oxygen consumed
 - (d) the mass of oxygen consumed.

- (a)(i) Write a balanced equation:



- (ii) Identify unknown and relate it to the quantity which is known:

$$\begin{aligned} n(\text{H}_2\text{O}) &= n(\text{H}_2) \\ &= 0.480 \text{ mol} \end{aligned}$$

- (b) Convert the moles to mass ($m=nM$):

$$\begin{aligned} n(\text{H}_2\text{O}) &= 0.480 \\ m(\text{H}_2\text{O}) &= 0.480 (18.0) \quad M(\text{H}_2\text{O}) = 18.0 \text{ g mol}^{-1} \\ &= 8.60 \text{ g} \end{aligned}$$

- (c) Relate unknown to known (using the equation):

$$\begin{aligned} n(\text{O}_2) &= \frac{1}{2} \times n(\text{H}_2) \\ &= \left(\frac{1}{2}\right) (0.480) \\ &= 0.240 \text{ mol} \end{aligned}$$

(d) Convert moles to mass ($m = nM$):

$$n(\text{O}_2) = 0.240$$

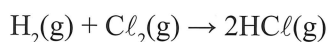
$$\begin{aligned} m(\text{O}_2) &= (0.240)(32.0) & M(\text{O}_2) &= 32.0 \text{ g mol}^{-1} \\ &= 7.68 \text{ g} \end{aligned}$$

2. 17.8 g of chlorine gas reacts with hydrogen to form hydrogen chloride gas.

Calculate:

- (a) the number of moles of hydrogen chloride formed and;
 (b) the mass of hydrogen chloride formed.

(a)(i) Write a balanced equation:



(ii) Identify the unknown and relate it to the quantity which is known:

$$\begin{aligned} n(\text{HCl}) &= 2 \times n(\text{Cl}_2) \\ &= 2 \times \frac{17.8}{70.9} & M(\text{Cl}_2) &= 70.9 \text{ g mol}^{-1} \end{aligned}$$

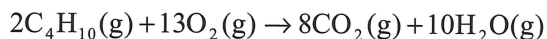
$$n(\text{HCl}) = 0.502 \text{ mol}$$

(b) Convert the moles to mass

$$\begin{aligned} n(\text{HCl}) &= 0.502 \\ m(\text{HCl}) &= (0.502)(36.458) & M(\text{HCl}) &= 36.458 \text{ g mol}^{-1} \\ &= 18.3 \text{ g} \end{aligned}$$

Set 29: Exercises

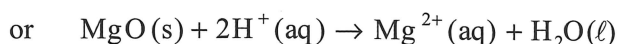
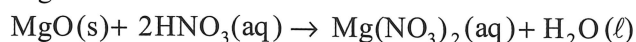
1. The equation for the combustion of butane is:



Calculate the number of moles of:

- (a) CO_2 produced in the combustion of 1.00 moles of C_4H_{10}
 (b) H_2O produced in the combustion of 3.00 moles of C_4H_{10}
 (c) O_2 consumed in the combustion of 0.600 moles of C_4H_{10}

2. 0.0300 moles of magnesium oxide is completely dissolved in nitric acid to form a solution of magnesium nitrate:



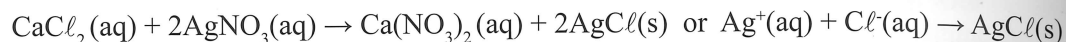
Calculate the number of moles and masses of:

- (a) nitric acid required
 (b) magnesium nitrate formed

Set 29: Stoichiometry

Notes

3. When silver nitrate solution is added to a solution of calcium chloride, a white precipitate of silver chloride is produced



If 0.200 moles of silver chloride is formed, calculate the number of moles and masses of :

- silver nitrate required
 - calcium chloride required
 - calcium nitrate formed in solution
4. A sample of copper(II) oxide was dissolved in sulfuric acid and the solution evaporated to dryness to yield 3.14 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$:
- $$\text{CuO}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) + 4\text{H}_2\text{O}(\ell) \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$$
- Calculate the mass of sulfuric acid required.
 - Calculate the moles of copper(II) oxide dissolved.
5. Write a balanced equation for the decomposition of potassium chlorate (KClO_3) into potassium chloride and oxygen gas by heating. How many moles of oxygen would be formed from the decomposition of 0.800 mol of KClO_3 ?
6. Write a balanced equation for the decomposition of mercury(II) oxide into mercury and oxygen gas (O_2). What mass of oxygen would be formed from the decomposition of 0.240 mol of mercury(II) oxide?
7. Excess nitric acid was added to 3.00 g of calcium carbonate:
- $$\text{CaCO}_3(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow \text{Ca}(\text{NO}_3)_2(\text{aq}) + \text{H}_2\text{O}(\ell) + \text{CO}_2(\text{g})$$
- or $\text{CaCO}_3(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + \text{H}_2\text{O}(\ell) + \text{CO}_2(\text{g})$
- Calculate the masses of:
- nitric acid consumed
 - carbon dioxide produced
 - calcium nitrate formed in solution
8. In the processing of uranium, one of the steps involves converting UO_2 to UF_6 .
- $$\text{UO}_2(\text{s}) + 4\text{HF}(\text{g}) + \text{F}_2(\text{g}) \rightarrow \text{UF}_6(\text{g}) + 2\text{H}_2\text{O}(\ell)$$
- For 7.50 kg of UO_2 , calculate:
- the mass of hydrogen fluoride required
 - the mass of fluorine required
 - the mass of UF_6 produced
9. A load of quarried limestone contains 92.0% CaCO_3 , the remainder being silica (SiO_2). The limestone is to be heated to form quicklime (CaO).
- $$\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$$
- What mass of quicklime could be produced by heating 5.00×10^2 kg of the limestone?
 - What mass of carbon dioxide would be produced?